

CognaSync

Clinical safety architecture — the four non-negotiable rules and the symptom-pattern logic

The defining design constraint of CognaSync is that every AI output respects a clear boundary: the system describes patterns in patient data; the clinician interprets what those patterns mean. The four rules below are enforced in three places — the system prompt, the structured data passed to the model, and a post-generation sanitization pass.

- 01 Never diagnose**
 No identification, detection, suggestion, or implication of any medical condition, mental health disorder, or psychiatric diagnosis. The rule applies to indirect framings ("this pattern is often associated with...") as well as explicit ones.
- 02 Never advise medication changes**
 No suggestion to start, stop, adjust the dose of, adjust the timing of, or substitute any medication. Observations that bear on medication route to provider discussion topics, never to patient action prompts.
- 03 Describe data, not clinical meaning**
 Observations are expressed as what the numbers show, not what they mean. "Mood averaged 4.2/10 over 14 days, trending down" is permitted; "mood has been consistently low" is not. Specificity protects against both overclaiming and hallucination.
- 04 Route, don't engage, in crisis**
 Crisis signals — explicit statements of self-harm or suicidal ideation — interrupt all analysis. The system replaces output with static crisis resources (988, Crisis Text Line, 911). The model never processes, comments on, or acknowledges the content of the triggering statement.

HOW THE RULES ARE ENFORCED

The system prompt instructs the model to follow all four rules at generation time. After generation, every output is post-processed against a forbidden-language list that catches diagnostic phrasing ("you have...", "you suffer from"), medication recommendations ("stop taking," "increase your dose"), and certainty overclaims ("this confirms," "this indicates [disorder]"). All quantitative scores — stability, stim load, sleep disruption, crash risk, nervous system load — are computed deterministically in Python before any AI call, so the model never invents or recomputes numbers; it only references values it receives as structured input.

SYMPTOM PATTERN DETECTION

Patients log symptoms (headache, fatigue, brain fog, nausea, dizziness, and others) inside the daily check-in. When a symptom appears on three or more days within a 60-day window, the system computes its co-occurrence with eleven tracked variables: sleep disruption, stim load, crash risk, mood, stress, energy, exercise minutes, alcohol units, hydration, workload friction, and perceived stress. A co-occurrence is surfaced only when the absolute difference between symptom-day and non-symptom-day means exceeds 1.5 on the variable's scale, and only when at least three matched observations support the comparison.

When a symptom meets reporting threshold, the system also queries medication events for any new prescription, dose change, or discontinuation within ± 14 days of the symptom's first appearance. If a change is found, it is reported as **temporal context, never as cause** — phrased as "these entries began around the same time as a change in [medication name] — worth mentioning to your provider," never as "this is consistent with [a withdrawal effect]" or "this is caused by [the medication change]."

The line is deliberate: the system surfaces patterns, the provider draws conclusions. No symptom interpretation. No causal claim. No recommendation. The provider receives the pattern with the data anchors needed to evaluate it clinically.